APT-BENCH: PATIENT-DISJOINT HIERARCHICAL CELL TYPING AND MEASUREMENT-EFFICIENT DISEASE RECOGNITION FROM SINGLE-CELL APTAMER PROFILES

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ABSTRACT

Single-cell assays produce hundreds of thousands of cells, but the effective sample size for clinical prediction is determined by independent subjects. We introduce APT-Bench, a patient-disjoint benchmark of 361,792 single-cell aptamer profiles from 40 patients across six disease groups. It separates five-lineage and 27-subtype cell annotation from patient-level disease recognition and measurement-efficiency analysis. Across neural and classical models, coarse lineage recognition is consistently easier than fine subtype prediction. Subtype F1 is more strongly associated with cell abundance than patient coverage, while most errors cross lineage boundaries, exposing a persistent granularity gap. Aggregated out-of-fold cell predictions correctly classify all 40 patients by disease, but patient-wise centering and identity-prediction audits reveal strong patient-specific absolute-expression signatures that cannot be separated into biological and technical components with the available metadata. Under nested patient-level selection, every outer fold chooses $B^* = 10$; the resulting 10-aptamer panels match the observed within-cohort performance of the full 293-aptamer panel. Random-panel and cross-model attribution controls indicate that this discriminative signal is distributed across alternative feature subsets. APT-Bench provides a statistically explicit setting for multiresolution, measurement-efficient single-cell learning, while limiting its disease conclusions to this cohort pending external validation.

1 INTRODUCTION

Cancer screening and diagnosis are usually framed at the patient level: does a person have disease, and if so which disease category is most likely? Single-cell liquid biopsy changes this framing. A single blood draw can support both a patient-level diagnostic decision and a cell-level characterization of the circulating immune compartment. These outputs are related, but they do not live at the same observational unit. Disease is a property of the patient, whereas lineage and subtype are properties of the cell.

This mismatch matters for evaluation. Our cohort contains more than 370,000 single cells, but only 40 patients. Treating cells as independent disease observations would substantially overstate the effective sample size of the diagnostic task. It would also hide a second asymmetry: lineage and subtype form a genuine cell-level hierarchy, but disease does not sit above that hierarchy. The clinically relevant problem is therefore not a single three-level taxonomy. It is a multi-resolution setting in which different targets require different evaluation protocols, inductive biases, and interpretations.

We study this setting on aptamer-based single-cell protein profiling. Aptamer (APT) measurements provide a fixed 293-dimensional panel of protein-target binding signals for each cell, offering a minimally invasive view of both disease-associated molecular state and immune composition from peripheral blood (Wu et al., 2025). This modality is attractive for repeated testing and potentially broad deployment, but it raises three intertwined modeling questions. First, how much predictive information is present at coarse versus fine biological resolution? Second, how sensitive is patient-level disease recognition to the mismatch between hundreds of thousands of correlated cells and only 40 independent patients? Third, how many aptamers and cells are needed to preserve disease recognition, and is that signal concentrated in a small unique panel or distributed across the assay?

We address these questions with APT-Bench, a patient-disjoint benchmark for multi-resolution characterization from single-cell APT profiles. The benchmark is organized around three primary tracks: cell-level coarse-to-fine hierarchy prediction, patient-level disease recognition with statistical-unit and confounding audits, and measurement-efficiency analysis over aptamer and cell budgets. This framing emphasizes that the central unresolved cell-level problem is not disease classification, which is already nearly saturated within cohort by simple baselines, but reliable hierarchical annotation under patient-independent evaluation.

To instantiate the hierarchy track, we use DropCascade, a differentiable lineage-subtype cascade. The model combines a shared APT encoder, soft lineage routing, and subtype experts. We compare it with classical models and encoder-matched flat and HCE-trained Transformer controls. It serves as a hierarchy-aware neural reference rather than a claimed universal winner; the matched controls test whether any gain survives control of encoder capacity and training budget.

This perspective leads to three empirical findings. First, coarse lineage recognition is substantially easier than fine subtype recognition across all models. A subtype-level audit shows that raw cell abundance explains more performance variation than patient coverage, but only partially, and that most wrong subtype predictions cross the true lineage rather than remaining among biological siblings. Second, disease labels are perfectly recoverable under the prespecified within-cohort patient folds, but sensitivity analyses show that this signal depends strongly on patient-specific absolute-expression structure, making overclaiming inappropriate in the absence of acquisition-batch metadata. Third, the disease signal is measurement-efficient but distributed: nested selection preserves patient-level performance with 10 aptamers, including in an LR sensitivity analysis over 50 repeated subsamples at 100 cells per patient, while 100-panel random controls and an XGBoost-SHAP attribution control argue against a unique indispensable biomarker set.

Our contributions are therefore as follows.

- **APT-Bench, a patient-disjoint benchmark for single-cell APT liquid biopsy.** We formalize coarse lineage prediction, fine subtype prediction, patient-level disease recognition, statistical-unit audits, and measurement-efficiency analysis within a single evaluation setting over 40 patients and more than 370,000 cells.
- **A subtype-centered account of the granularity gap.** Beyond aggregate macro-F1, we quantify how cell abundance and patient coverage relate to each subtype's performance, decompose incorrect subtype predictions into sibling and cross-lineage errors, and report the coarse-to-fine gap for every model. We use DropCascade as a hierarchy-aware reference model rather than claiming universal superiority.
- **A disciplined disease-recognition and measurement-efficiency audit.** We show that disease recognition is extremely strong within cohort but sensitive to patient-specific absolute-expression structure, and that nested aptamer-panel selection plus equal-cell subsampling reveal a low measurement budget under the current within-cohort task.

2 Related Work

Liquid biopsy and multi-cancer detection. Blood-based multi-cancer early detection has focused primarily on circulating tumor DNA, methylation, fragmentomics, and bulk protein assays (Cohen et al., 2018; Zeng et al., 2026; Wan et al., 2025; Kahwati et al., 2025). These approaches target a patient-level diagnostic decision. Our setting differs because APT profiling observes individual circulating cells, which introduces a second prediction problem at cell resolution: identifying lineage and fine subtype structure within the same blood draw. The resulting task is not simply "more labels," but a change in statistical unit and hierarchy structure.

Single-cell foundation models and modality mismatch. Large pretrained models such as Geneformer, scGPT, and scFoundation learn representations from transcriptomic data and are widely used as reference points for single-cell representation learning (Theodoris et al., 2023; Cui et al., 2024; Hao et al., 2024). Their success does not transfer automatically to our setting. The input modality is different, since APT consists of a fixed 293-dimensional protein-oriented panel rather than a gene-vocabulary sequence. More importantly, recent evaluation work shows that frozen or zero-shot single-cell foundation-model embeddings are often matched by simpler baselines once the downstream task and evaluation protocol are made explicit (Kedzierska et al., 2025). This motivates training directly on APT profiles and treating evaluation design itself as a first-class contribution.

Hierarchical cell-type classification. Cell identity naturally admits coarse-to-fine structure, and recent work has explored losses or architectures that exploit this structure for annotation and out-of-distribution robustness (Jia et al., 2025; Cultrera di Montesano et al., 2026). Our use of hierarchy is narrower and more explicit: disease is kept outside the hierarchy, while lineage and subtype form the only genuine taxonomy. We compare DropCascade not only with flat classifiers but also with a matched flat FT-style Transformer control (Gorishniy et al., 2021) and the same control trained with a two-level adaptation of hierarchical cross-entropy (HCE), the most directly related hierarchy-aware loss.

Measurement-efficient disease recognition. Clinical translation of high-plex assays depends not only on accuracy but also on measurement cost. APT-Bench therefore includes a compact-panel analysis that asks how many aptamers are needed to preserve patient-level disease recognition under nested patient-disjoint evaluation. This is distinct from biomarker discovery: our goal is to quantify predictive budget and redundancy in the assay, not to claim that a small panel is a universal causal signature.

Tabular deep learning versus classical machine learning. APT expression is a medium-dimensional tabular signal. In that regime, gradient-boosted trees remain strong defaults and often rival or outperform deep tabular models (Grinsztajn et al., 2022; Erickson et al., 2026). This literature is directly reflected in our results: XGBoost and encoder-matched Transformer controls remain competitive at both resolutions, so DropCascade cannot be justified by a blanket accuracy claim. We instead use it as a structured reference for the hierarchy diagnostics, while disease recognition is audited with simple, strong tabular baselines.

3 Problem Setup

3.1 Dataset and Prediction Targets

Our dataset contains single-cell aptamer (APT) protein expression profiles from peripheral blood collected from 40 patients across six disease groups: colorectal cancer (CRC), gastric cancer (GC), hepatocellular carcinoma (HCC), pancreatic cancer (PC), biliary tract cancer (BTC), and healthy controls (Normal), with 6–8 patients per group. The raw cohort contains 374,854 cells. After excluding the transcriptome-derived *Unknown* subtype label, which denotes uncertain annotation rather than a validated biological state, the main benchmark contains 361,792 cells. Each cell is represented by a 293-dimensional APT expression vector.

The 293 aptamers correspond to designed protein-target probes rather than an arbitrary learned basis. Expression values are log-transformed and standardized feature-wise using statistics estimated from the training partition only. Lineage and subtype labels are obtained from paired transcriptomic profiling on the same cells through barcode matching. The transcriptomic modality is used only to define supervision and is never provided as input to the models.

APT-Bench contains two kinds of targets at different observational units.

- **Patient-level disease recognition.** Each patient is assigned to one of six disease groups.
- **Cell-level hierarchy prediction.** Each cell is assigned to one of five coarse immune lineages and one of 27 fine subtypes. Every subtype maps to exactly one parent lineage through a manually curated lookup table.

Disease is therefore not treated as the root of a three-level hierarchy. It is a separate patient-level label observed on the same blood draw.

3.2 Evaluation Protocols

We evaluate models under two main protocols with different purposes.

Patient-level 5-fold cross-validation. This is the primary protocol for all headline claims. Patients are partitioned into five disease-stratified folds, each containing eight held-out test patients. At every fold, preprocessing statistics, model fitting, and hyperparameter selection use only the training and validation patients from the remaining folds. We pool out-of-fold predictions across all five test folds, yielding a patient-disjoint evaluation spanning all 40 patients.

Cell-level split control. As a protocol-sensitivity control, cells are randomly split irrespective of patient identity. Cells from the same patient can therefore appear in both training and evaluation sets. We do not treat this as a realistic deployment protocol. Its only role is to quantify performance inflation caused by leakage.

3.3 Class Distribution

Because the subtype distribution is long-tailed, we use macro-F1 as the primary metric for coarse-lineage and fine-subtype prediction. Accuracy is reported as a secondary metric. For uncertainty quantification under the main patient-level protocol, we compute 95% confidence intervals from 2,000 paired patient-level bootstrap resamples over the 40 pooled test patients. The primary inferential family contains DropCascade versus HCE and versus XGBoost at the coarse and fine endpoints. Two-sided bootstrap p-values use twice the smaller tail probability with a plus-one correction and are Holm-adjusted across these four tests; comparisons outside this family are exploratory.

From the same pooled predictions, we also report per-subtype support and F1, support–performance rank correlations, sibling versus cross-lineage errors, and each model's coarse-minus-fine gap as descriptive difficulty diagnostics rather than new ranking metrics.

We report disease recognition at the patient level, since the clinical decision is made per patient rather than per cell. Cell-level disease accuracies are included only as supplementary evidence and are not interpreted as though hundreds of thousands of cells were independent disease observations.

4 Hierarchy-Aware Reference Model

APT-Bench is intentionally broader than any single model. We nevertheless need a hierarchy-aware reference architecture that can operate directly on APT profiles under the benchmark protocols. DropCascade plays this role. It focuses on the genuine cell-level hierarchy, lineage $\rightarrow$ subtype, and leaves disease recognition to separate tabular baselines and patient-level aggregation. We do not present the architecture as a primary contribution: without full five-fold component ablations, its global path, coarse auxiliary objective, soft routing, and contrastive construction should be read as implementation choices rather than independently validated sources of performance.

4.1 Cell Encoder

Each cell is represented by a standardized APT vector $x \in \mathbb{R}^{293}$. We embed feature identity and feature value separately. For aptamer index i, we form a token

$$t_i = \mathrm{Emb}(i) + \mathrm{MLP}_{\mathrm{val}}(x_i), \qquad i = 1, \ldots, 293. \tag{1}$$

A learnable `[CLS]` token is prepended to the resulting sequence and processed by a pre-norm Transformer encoder with 4 layers, 8 attention heads, hidden dimension 256, feed-forward dimension 512, GELU activations, and dropout 0.1. The final `[CLS]` representation

$$z = \mathrm{Encode}(x) \in \mathbb{R}^{256} \tag{2}$$

serves as the shared cell representation for downstream tasks.

4.2 SOFT-ROUTED HIERARCHICAL CASCADE

Let $M = 5$ be the number of coarse lineages and $K = 27$ the number of fine subtypes. Each subtype k belongs to exactly one parent lineage $m(k)$. A coarse head produces lineage logits

$$c = W_c z \in \mathbb{R}^M, \tag{3}$$

with routing distribution

$$p(m \mid x) = \mathrm{softmax}(c/\tau)_m, \tag{4}$$

where $\tau = 1$.

Instead of a hard cascade, which would force subtype prediction to follow only the top predicted lineage, DropCASCADE uses a soft mixture of a global subtype path and lineage-specific experts:

$$s_k(x) = \underbrace{g_k(z)}_{\text{global path}} + \alpha\, p(m(k) \mid x)\, e_{m(k),k}(z) + \gamma \log\big(\epsilon + (1-\epsilon) p(m(k) \mid x)\big), \tag{5}$$

Here $g(z)$ is a global subtype head, $e_{m,k}(z)$ is the expert associated with lineage m, $\alpha = 1.0$, $\epsilon = 0.1$, and γ is ramped from 0 to 0.5 during early training. The global path retains a direct subtype-logit contribution alongside the routed expert contribution; we do not isolate its effect from the other components. Final predictions are

$$\hat{k} = \arg\max_k s_k(x), \qquad \hat{m} = \arg\max_m c_m. \tag{6}$$

4.3 CROSS-DISEASE CONTRASTIVE REGULARIZATION

The reference model includes a cross-disease supervised contrastive term. Let

$$u = \frac{\mathrm{Proj}(z)}{\|\mathrm{Proj}(z)\|_2} \tag{7}$$

be a normalized projection of the encoder representation. For anchor cell i, positives are cells with the same subtype but a different disease label. Same-subtype cells from the same disease are excluded by construction. Cells from different subtypes are treated as negatives, with additional emphasis on negatives that share the same coarse lineage. The contrastive loss weight is ramped from 0 to 0.2 after the early warmup period. Because we do not compare this construction with standard supervised contrastive learning across all five folds, we make no claim that the cross-disease positive rule independently improves generalization.

4.4 OPTIMIZATION AND MODEL SCOPE

The classification cascade is trained jointly using

$$\mathcal{L} = \lambda_{\text{fine}} \mathcal{L}_{\text{fine}} + \lambda_{\text{coarse}} \mathcal{L}_{\text{coarse}} + w_{\text{contrast}}(t) \mathcal{L}_{\text{contrast}}, \tag{8}$$

where $\mathcal{L}_{\text{fine}}$ and $\mathcal{L}_{\text{coarse}}$ are cross-entropy losses with $\lambda_{\text{fine}} = 1.0$ and $\lambda_{\text{coarse}} = 0.5$. The contrastive weight increases to 0.2 according to the schedule in §4.3.

We optimize the cascade using AdamW with learning rate 10^{-4} and weight decay 10^{-5}, gradient clipping at norm 1.0, batch size 512, and mixed-precision training. Models are trained for at most 50 epochs with early stopping based on validation fine-level macro-F1 with patience 10. The final configuration does not use class-balanced sampling, as our experiments found that it did not improve performance after correcting an implementation issue; additional details are provided in Appendix A.8.

Matched Transformer controls. To isolate encoder capacity from hierarchical modeling, a flat FT-style Transformer control (Gorishniy et al., 2021) removes the cascade while retaining the identical APT tokenizer, Transformer dimensions, optimizer, early stopping, and training budget above. We call it FT-style because its tokenizer is exactly the APT feature tokenizer used by DropCASCADE, rather than the original numerical/categorical FT-Transformer tokenizer. This model and its HCE counterpart predict the same $M + K$ lineage-and-subtype nodes. The flat model applies independent cross-entropy to the lineage and subtype logit slices.

Aggregation / Representation	Patient Acc.	Patient Macro-F1	Macro-AUROC	95% CI (Acc.)
Cell-level classifiers*, majority vote	**1.000**	**1.000**	–	[0.912, 1.000]
Patient-level APT median + LR†	0.950	0.948	1.000	–
Gold subtype composition (CLR) + LR‡	0.575	0.572	0.842	–
Gold lineage composition (CLR) + LR‡	0.400	0.377	0.678	–

* All five cell-level classifiers (Logistic Regression, XGBoost, Linear SVM, Random Forest, Reference Correlation) reach the identical $40/40$ majority-vote result and are therefore collapsed into a single row; see Table 4 for their (distinct) cell-level accuracies. † Genuinely patient-level feature (median APT expression per patient per aptamer, $n = 40$, no cell-level pseudo-replication). ‡ Gold-label immune-cell composition (paired transcriptomic annotation, not model-predicted subtypes); CLR = centered log-ratio transform of the compositional proportions. All rows use the same patient-level 5-fold partition (§3.2); the confusion matrix for the majority-vote rows is diagonal ($40/40$ patients correctly assigned; see Appendix A.1).

Table 1: **Patient-level** disease classification ($n = 40$ patients), the correct statistical unit for a per-patient diagnostic call. All five cell-level classifiers agree perfectly once cell-level predictions are aggregated to the patient by majority vote (95% Clopper-Pearson CI on the $40/40$ result: $[0.912, 1.000]$). A model trained on purely patient-level summary features, with no cell-level pseudo-replication at all, achieves comparably strong discrimination, and immune-cell composition alone (gold-label, CLR-transformed) is also clearly above chance. Because the cohort contains only 40 patients (6–8 per disease) and no acquisition-batch metadata (plate, run, or date) exists for this dataset, we cannot exclude a disease-correlated technical contribution to this performance; see Appendix A.1 for the sensitivity analyses (patient-wise centering, patient-fingerprint diagnostic) that motivate this caveat, and §6 for a full discussion. This result should be read as within-cohort feasibility, not evidence of clinical generalization.

For HCE (Cultrera di Montesano et al., 2026), let $R_{ij} = 1$ when node j is node i or its descendant, $p = \mathrm{softmax}(q)$ over all nodes, and $s = Rp$. We minimize $-\lambda_{\text{fine}} \log s_{M+k} - \lambda_{\text{coarse}} \log s_m$ for observed subtype k and lineage m. Because every fine label here is a leaf, the fine HCE term alone would reduce to ordinary CE; including the simultaneously observed parent label makes this a stated two-level adaptation rather than an unqualified reproduction of the mixed-granularity setting in the original work. All three Transformer models are evaluated from the checkpoint selected by patient-validation fine macro-F1, without an additional post-selection calibration stage.

Disease recognition is not handled by a dedicated head inside DropCascade. This is a deliberate modeling decision rather than a claim that disease is solved in a strong external sense: within-cohort disease prediction is already nearly saturated by simple classical baselines, while the architectural challenge addressed by DropCascade is the unresolved cell-level hierarchy problem.

5 Results

5.1 Disease Recognition Is Strong Within Cohort, but Requires Caveats

We begin with disease because it is the clinically salient target, then immediately separate it from the central hierarchy benchmark analysis. Under the patient-level 5-fold protocol, all five classical cell-level baselines yield the same patient-level majority-vote result: all 40 patients are correctly assigned, for a Clopper-Pearson 95% interval of $[0.912, 1.000]$ (Table 1). A genuinely patient-level model using each patient's median APT profile also performs strongly, reaching accuracy 0.950 and macro-AUROC 1.000. In 1,000 patient-label permutations, none matched its observed accuracy (plus-one $p = 0.000999$; Appendix A.1).

These results do not justify a claim of solved clinical diagnosis. Sensitivity analyses in Appendix A.1 show that the disease signal depends strongly on patient-specific absolute-expression structure: patient-wise centering reduces patient-level accuracy from 1.000 to 0.650, and a 40-way patient-identity diagnostic reaches 88.4% accuracy on held-out cells. Because acquisition-batch metadata are unavailable, the observed offsets cannot be cleanly decomposed into biological versus technical sources. We therefore use disease recognition primarily to motivate the need for careful statistical-unit accounting in APT-Bench, not as the main architectural success story.

Task	Model	Macro-F1	Accuracy	95% CI (Macro-F1)
Coarse lineage	DropCascade	0.326	0.502	[0.301, 0.352]
	Flat FT-style Transformer	0.323	0.507	[0.302, 0.345]
	FT-style Transformer + HCE	**0.333**$^{\text{n.s.}}$	0.503	[0.312, 0.354]
	XGBoost	0.322$^{\text{n.s.}}$	**0.517**	[0.300, 0.345]
Fine subtype	DropCascade	**0.152**	0.305	[0.129, 0.166]
	Flat FT-style Transformer	0.147	0.315	[0.121, 0.161]
	FT-style Transformer + HCE	0.151$^{\text{n.s.}}$	0.306	[0.128, 0.167]
	XGBoost	0.143$^{\text{n.s.}}$	**0.323**	[0.128, 0.158]

Table 2: Patient-disjoint 5-fold hierarchy results on pooled out-of-fold predictions from all 40 patients. The primary comparisons are DropCascade versus HCE and versus XGBoost, for coarse and fine macro-F1. Intervals and two-sided tests use the same 2,000 paired patient bootstrap resamples; the four primary test p-values are Holm-corrected. The matched Flat FT-style Transformer is an encoder-capacity diagnostic, and all other row-wise comparisons are exploratory. The two FT-style controls match DropCascade's tokenizer, encoder dimensions, optimizer, early stopping, training budget, and checkpoint rule; HCE is defined in §4.4. For primary comparisons, † denotes a significantly lower comparator, ‡ a significantly higher comparator, and $^{\text{n.s.}}$ no Holm-adjusted significance. Coarse–fine consistency is reported only in Appendix A.6.

5.2 Patient-Disjoint Cell-Level Hierarchy Prediction Remains Difficult

The harder unresolved problem is cell-level hierarchy prediction under patient-disjoint evaluation. Table 2 shows that coarse lineage recognition is consistently easier than fine subtype recognition. For DropCascade, coarse macro-F1 reaches 0.326 while fine macro-F1 is only 0.152; for the matched flat FT-style Transformer, the corresponding values are 0.323 and 0.147. This granularity gap appears across model families and indicates that APT captures broad immune identity more reliably than fine subtype structure.

The matched comparison changes the architectural conclusion. Relative to the flat Transformer, DropCascade differs by only $+0.0036$ coarse macro-F1 (paired patient-bootstrap 95% CI $[-0.0078, 0.0146]$) and $+0.0055$ fine macro-F1 (95% CI $[-0.0021, 0.0125]$). HCE has the highest coarse point estimate (0.333), but its difference from DropCascade is also unsupported: DropCascade$-$HCE is -0.0069 (95% CI $[-0.0154, 0.0026]$) at the coarse level and $+0.0015$ (95% CI $[-0.0083, 0.0047]$) at the fine level. Differences from XGBoost likewise do not exclude zero. Thus the experiment does not isolate a performance gain from the cascade: the defensible result is the persistent coarse-to-fine difficulty under patient-disjoint evaluation, not superiority of a routing mechanism.

A coarse–fine consistency diagnostic computed from these same pooled out-of-fold predictions is reported in Appendix A.6. We keep it as supporting evidence because consistency can be altered mechanically by constrained decoding and is not a standalone measure of predictive quality.

5.3 Why 361,792 Cells Still Yield Only 0.15 Subtype Macro-F1

Cell abundance explains more than patient coverage, but neither is sufficient. For subtype k, let n_k be its cell count, P_k its patient count, and $F_{1,k}$ its pooled out-of-fold F1. We compare $\rho_{\text{cell}} = \text{Spearman}(F_{1,k}, \log n_k)$ with $\rho_{\text{patient}} = \text{Spearman}(F_{1,k}, P_k)$. Table 3 shows that the former is larger for all eight models (0.486 versus 0.288 for DropCascade; 0.540 versus 0.275 for XGBoost). Patient coverage has limited resolution because 16 of 27 subtypes occur in all 40 patients. Abundance is still incomplete: tail subtype Plasma reaches F1 $\approx$ 0.61, whereas head subtype Memory B-1 remains near 0.03. Appendix A.2 reports all requested per-subtype quantities.

Most subtype errors cross lineage boundaries. We define $R_{\text{sibling}} = \Pr(m(\hat{k}) = m(k) \mid \hat{k} \neq k)$ using the fixed parent map $m(\cdot)$. Only 25.0% of DropCascade errors and 24.6% of XGBoost errors are sibling confusions; across all models, 70.5–80.9% of errors cross the true lineage (Table 3). Low fine-level F1 is therefore not primarily benign confusion among nearby siblings.

The granularity gap is a benchmark finding. For each model, $G = F_1^{\text{coarse}} - F_1^{\text{fine}}$ is positive and ranges from 0.099 to 0.200. We use G only to expose persistent coarse-to-fine difficulty, not as a new metric contribution.

Model	$\boldsymbol{\rho}_{\text{cell}}$	$\boldsymbol{\rho}_{\text{patient}}$	G	R_{sibling}	**Cross-lineage**
DropCascade	0.486	0.288	0.174	0.250	0.750
Flat FT-style Transformer	0.505	0.255	0.176	0.243	0.757
FT-style Transformer + HCE	0.513	0.318	0.183	0.257	0.743
XGBoost	0.540	0.275	0.179	0.246	0.754
Logistic Regression	0.435	0.107	0.167	0.295	0.705
Linear SVM	0.339	-0.012	0.200	0.265	0.735
Random Forest	0.661	0.418	0.158	0.191	0.809
Reference Correlation	0.376	0.003	0.099	0.225	0.775

Table 3: Subtype support, granularity, and error diagnostics on pooled patient-disjoint predictions. The two ρ columns are Spearman associations of per-subtype F1 with log cell count and patient coverage. $G = F_1^{\text{coarse}} - F_1^{\text{fine}}$. Among incorrect subtype predictions, R_{sibling} retains the true parent lineage and Cross-lineage is its complement. These are descriptive benchmark findings, not proposed ranking metrics.

5.4 Protocol Sensitivity: Patient-Level versus Cell-Level Splitting

The cell-level split control (Appendix A.3) makes both DropCascade and XGBoost look stronger, but the inflation is larger for DropCascade: +0.090 versus +0.050 on coarse lineage, and +0.108 versus +0.069 on fine subtype. Leakage can therefore change not only absolute performance but also the apparent advantage of a method family. APT-Bench treats patient-disjoint evaluation as a non-negotiable requirement, not a reporting detail.

5.5 Disease Signal Is Measurement-Efficient but Predictively Distributed

The compact-panel analysis in Appendix A.7 provides a final perspective on what the APT modality supports. Under a nested patient-level selection pipeline, all five outer folds choose the same development tolerance budget, $B^* = 10$, and the resulting 10-aptamer panels reproduce the full-panel patient-level macro-F1 of 1.0 for both Logistic Regression and XGBoost on every outer test fold. An LR sensitivity analysis obtains the same macro-F1 for all 50 equal-cell subsamples at 100 cells per patient. Across 100 Random-B panels per compact budget, performance also rises quickly with B; moreover, LR-coefficient and XGBoost-SHAP Top-10 panels overlap only moderately but both retain perfect patient-level performance. The correct interpretation is therefore not that the dataset isolates a unique minimal biomarker set, but that within-cohort disease information is highly redundant and can be recovered with substantially reduced aptamer and cell-count budgets.

Compact-panel recovery does not resolve whether the disease signal is biological, technical, or mixed. It instead shows a central asymmetry: within-cohort disease recognition is statistically and operationally easier than patient-disjoint cell-level hierarchy prediction.

6 Discussion

APT-Bench shows that multi-cancer liquid biopsy decomposes into targets with different statistical units and evidentiary standards. Across matched models, coarse lineage prediction is easier than fine subtype prediction; abundance explains more subtype variation than patient coverage, yet most errors still cross lineage boundaries. Disease recognition is strong within cohort but vulnerable to patient-specific absolute-expression effects. Compact aptamer and cell budgets preserve this signal, indicating measurement efficiency rather than unique causal biomarkers.

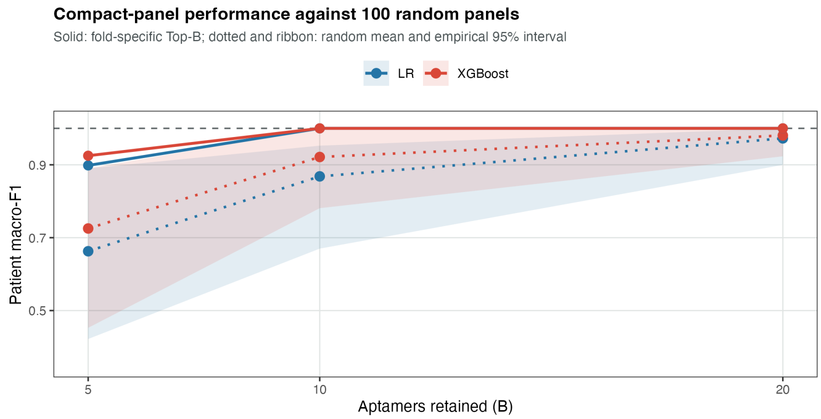

Figure 1: Patient-level disease macro-F1 under compact aptamer-budget controls. Solid lines show fold-specific Top-B panels; dotted lines and ribbons show the mean and empirical 95% interval over 100 independently sampled Random-B panels. The dashed line is the full 293-aptamer reference. Scores pool the 40 outer-test patients, each evaluated exactly once.

Matched controls show no significant DropCascade advantage at either resolution; persistent cross-lineage errors instead point toward better patient-independent representations.

6.1 Limitations

Disease-level novelty is exploratory. The appendix leave-one-disease-out stress test has too few disease groups and too little proxy-novel diversity to support a main claim.

The disease result is within-cohort and potentially confounded. Only 40 patients are available and acquisition-batch metadata are missing; patient fingerprints and centering sensitivity show that biological and technical offsets cannot be separated.

Matched baselines narrow the method claim. The flat and HCE controls prevent attributing Transformer capacity to the cascade. They do not support a DropCascade-specific gain, so it remains a reference implementation rather than a superior model.

Compact panels imply measurement efficiency, not unique biomarkers. Small-panel and equal-cell analyses show distributed predictive signal and lower measurement budgets, not a universal causal biomarker set.

Future work requires external cohorts with batch metadata and out-of-cohort validation of subtype models and compact panels.

7 Conclusion

APT-Bench separates patient-disjoint hierarchy typing, patient-level disease audit, and measurement efficiency. Disease recognition remains strong under compact budgets but is sensitive to patient-specific structure, while subtype typing remains difficult despite 361,792 cells. Matched Transformer and HCE controls do not support a DropCascade-specific gain; the persistent granularity gap and cross-lineage errors instead motivate more patient-independent representations.

AI use statement

TODO: **Required for ICLR 2027.** Disclose every generative-AI tool and model used for writing, editing, coding, analysis, literature review, or figure preparation; identify the tasks for which each tool was used, describe human verification of AI-assisted outputs, and state that the authors take responsibility for the final text, claims, code, and artifacts.

Ethics statement

TODO: State the human-subjects approval or exemption and informed-consent status; describe data de-identification, access restrictions, privacy protections, intended clinical/research use, foreseeable misuse or bias risks, and any conflicts of interest. Do not include identifying institutional details if they would break anonymity.

Reproducibility statement

TODO: Provide the anonymous code and data-access locations, exact patient-split manifest, preprocessing and evaluation commands, software environment, random seeds, model hyperparameters, compute requirements, and paths for the prediction artifacts and tables used in every reported result.

Author Contributions

TODO: Add author contributions using CRediT roles (for example: conceptualization, methodology, software, validation, formal analysis, data curation, visualization, writing, supervision, and funding acquisition). Preserve anonymity in the review version if required.

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A Appendix

A.1 Disease Classification: Cell-Level Source Prediction and Patient-Level Sensitivity Analyses

Cell-level prediction of the disease source of a held-out patient. Table 1 reports patient-level disease classification, the correct statistical unit for a per-patient diagnostic call. For reference and comparability with earlier reporting conventions, Table 4 additionally reports the pooled *cell-level* metric, in which every held-out cell is treated as an independent prediction target and pooled across all 361,792 cells from the five test folds. This is an easier and statistically different task than patient-level diagnosis, since cells from the same patient are correlated observations rather than independent draws and patients with more cells contribute more weight. We therefore report this metric only as a lower-level diagnostic.

Patient-wise centering sensitivity. To test whether disease prediction relies on each patient's absolute expression level rather than only on within-patient relative structure, we repeated Logistic Regression training and evaluation after subtracting each patient's own per-aptamer median from every cell in both training and test data. This transformation preserves within-patient cell-to-cell variation while removing a patient-specific location shift. Cell-level accuracy drops from 0.991 to 0.401 (macro-F1 $0.989 \rightarrow 0.348$), and patient-level majority-vote accuracy drops from $40/40$ to $26/40$. This collapse shows that disease recognition depends substantially on patient-level absolute-expression structure, but it does not by itself determine whether that structure is biological, technical, or a mixture of both.

Model	Accuracy	Macro-F1	Weighted-F1	Balanced Acc.
Logistic Regression	**0.991**	**0.989**	**0.991**	**0.989**
XGBoost	0.990	0.988	0.990	0.987
Linear SVM	0.988	0.986	0.988	0.985
Random Forest	0.963	0.955	0.963	0.953
Reference Correlation†	0.921	0.909	0.921	0.915

† SingleR-style reference-correlation classifier (Spearman nearest-centroid mapping), following the template-matching principle used by reference-mapping methods such as SingleR (Aran et al., 2019).

Table 4: **Cell-level** prediction of the disease source of a held-out patient's cells (6-class), pooled over all 361,792 out-of-fold cells from the patient-level 5-fold protocol. This is a different, easier statistical task than the patient-level diagnosis reported in Table 1: cells from the same patient are correlated observations rather than independent draws, and this metric implicitly weights patients with more cells more heavily. We report it here only as a cell-level reference point, not as a patient-level diagnostic result.

Patient-fingerprint diagnostic. We additionally trained a 40-way classifier to predict patient identity from individual cells, using a within-patient 50/50 cell split for this diagnostic only. The classifier identifies the correct patient from a single held-out cell with 88.4% accuracy (macro-F1 0.862), against a 2.5% chance level. Individual cells therefore carry a strong and easily recoverable patient-specific signature, consistent with the centering sensitivity above.

Patient-level summary-feature control. Finally, to confirm that the disease signal does not rely only on cell-level pseudo-replication, we trained a genuinely patient-level classifier using each patient's median 293-dimensional APT profile. This summary-feature model reaches patient accuracy 0.950, patient macro-F1 0.948, and macro-AUROC 1.000. Together with the sensitivity analyses above, this indicates that disease information is present at the patient level but strongly entangled with absolute-expression offsets that cannot be cleanly separated without acquisition-batch metadata.

Patient-label permutation control. We evaluated the same patient-level median-profile classifier under 1,000 label permutations. Disease labels were permuted across the 40 patients as indivisible units, preserving the disease-group counts $(6, 8, 7, 7, 6, 6)$; labels were never shuffled across cells. We retained the original five patient folds rather than restratifying after permutation. No permuted dataset reached the observed accuracy of 0.950 or macro-F1 of 0.948, giving plus-one permutation $p = (0 + 1)/(1000 + 1) = 0.000999$ for both statistics. The null accuracy mean was 0.137, with empirical 95% interval $[0.025, 0.275]$. This rejects a random patient-label explanation under the fixed protocol, but does not distinguish biological signal from patient-associated technical structure.

A.2 Per-Subtype Support and Performance Audit

Table 5 gives the complete subtype-level quantities underlying the coverage analysis in §5.3. Counts and patient coverage are computed from the pooled ground-truth labels, while F1 is computed from the same patient-disjoint out-of-fold predictions used for the main results. Mean cells per patient averages only over patients in which the subtype is observed. The table includes both DropCascade and XGBoost to show that the main abundance-versus-coverage pattern is not specific to one architecture.

Table 5: Per-subtype support and pooled out-of-fold F1. Cells/patient is the mean among patients in which that subtype is observed. Head and Tail denote the seven most and seven least abundant subtypes; all others are Mid.

Subtype	Lineage	Cells	Patients	Cells/patient	Regime	DropCascade F1	XGB F1
Erythrocyte	Other	65	6	10.8	Tail	0.253	0.133
Atypical Memory B	B	781	34	23.0	Tail	0.005	0.005
CD4 Monocyte-4	Myeloid	1,405	35	40.1	Tail	0.027	0.014
platelet	Other	1,574	38	41.4	Tail	0.078	0.063

Continued on next page

Subtype	Lineage	Cells	Patients	Cells/patient	Regime	DropCascade F1	XGB F1
Tcm	T	2,121	36	58.9	Tail	0.000	0.004
Memory B-2	B	2,130	33	64.5	Tail	0.017	0.011
Plasma	B	2,173	40	54.3	Tail	0.611	0.599
pDC	Myeloid	2,615	40	65.4	Mid	0.029	0.019
CD14 Monocyte-3	Myeloid	2,977	32	93.0	Mid	0.258	0.177
CD8 + CTL-3	T	3,244	27	120.1	Mid	0.020	0.023
GCB	B	3,283	31	105.9	Mid	0.039	0.047
Cycling T	T	3,667	40	91.7	Mid	0.028	0.012
CD14 Monocyte-2	Myeloid	3,931	40	98.3	Mid	0.075	0.078
CD8 + CTL-2	T	4,049	33	122.7	Mid	0.069	0.078
CD16 Monocyte-2	Myeloid	5,142	40	128.6	Mid	0.111	0.117
CD8 + CTL-1	T	7,303	40	182.6	Mid	0.009	0.002
cDC2	Myeloid	7,376	40	184.4	Mid	0.132	0.125
CD8 Tem-2	T	10,548	40	263.7	Mid	0.019	0.017
CD16 Monocyte-1	Myeloid	14,678	40	366.9	Mid	0.152	0.118
CD4 + Tcm-2	T	16,861	39	432.3	Mid	0.188	0.188
Naive T	T	17,960	40	449.0	Head	0.131	0.149
Memory B-1	B	23,722	40	593.0	Head	0.034	0.032
CD8 Tem-1	T	26,714	40	667.9	Head	0.187	0.162
CD14 Monocyte-1	Myeloid	29,871	40	746.8	Head	0.437	0.451
CD56 NK	NK	38,442	40	961.0	Head	0.398	0.437
CD16 NK	NK	58,507	40	1462.7	Head	0.383	0.376
CD4 + Tcm-1	T	70,653	40	1766.3	Head	0.417	0.435

For continuity with common long-tail reporting, Table 6 aggregates the seven least and seven most abundant subtypes. This summary is secondary to the per-class audit because individual subtypes can depart sharply from the abundance trend.

Model	Tail Macro-F1	Head Macro-F1
DropCascade	0.142	0.284
Flat FT-style Transformer	0.117	0.290
FT-style Transformer + HCE	0.125	0.288
XGBoost	0.119	0.292
Logistic Regression	0.114	0.197
Linear SVM	0.082	0.204
Random Forest	0.072	0.245
Reference Correlation	0.060	0.114

Table 6: Fine-subtype macro-F1 on the seven least and seven most abundant subtypes under pooled patient-disjoint evaluation. This aggregate is reported as a descriptive complement to the complete per-subtype audit in Table 5.

A.3 Protocol Sensitivity: Patient-Level versus Cell-Level Splitting

Table 7 compares the primary patient-disjoint protocol with a leakage-prone cell-level split control. It is placed in the appendix because the central result uses patient-disjoint evaluation throughout; the control quantifies how much the easier protocol would inflate both absolute performance and the apparent advantage of a model family.

Task	Model	Patient-disjoint	Cell split	Inflation
Coarse lineage	XGBoost	0.322	0.372	+0.050
	DropCascade	0.326	0.417	+0.090
Fine subtype	XGBoost	0.143	0.212	+0.069
	DropCascade	0.152	0.260	+0.108

Table 7: Macro-F1 under the primary patient-disjoint protocol versus a leakage-prone cell-level split control. Cell-level splitting inflates performance for both methods, and does so more strongly for DropCascade than for XGBoost. This demonstrates that evaluation protocol can change both absolute performance and the apparent relative advantage of model families, which is why APT-Bench treats patient-disjoint evaluation as a requirement rather than a reporting option.

A.4 Patient-Level Lineage-Resolved Attribution Audit

We revisit disease attribution at the patient, rather than cell, level. For patient p, lineage m, and aptamer j, we compute $\bar{x}_{pmj} = n_{pm}^{-1} \sum_{i \in (p,m)} x_{ij}$, so every point in Figure 2 represents one patient regardless of cell count. We show the three aptamers with the best mean rank across the five outer-fold PSAS rankings (APT-179, APT-225, and APT-164) in the three major immune compartments. The raw view is paired with a patient-centered view obtained by subtracting each patient's per-aptamer median across all of their cells before lineage aggregation. Disease groups contain only 6–8 patients, and the two views use the same vertical scale.

This is an exploratory full-cohort visualization, not an independently validated biomarker analysis. Raw patient means separate several disease groups, but much of that separation contracts toward zero after patient centering, consistent with the centering sensitivity in Appendix A.1. The available feature manifest contains only anonymized APT identifiers and no verified protein-target mapping. We therefore make no molecular-plausibility, protein-target, or pathway claim from these panels.

A.5 A Euclidean Heuristic Inspired by Hyperbolic Hierarchy Embedding

Motivated by the observation that tree-structured label hierarchies can be embedded with lower distortion under hyperbolic geometry than Euclidean geometry (Nickel & Kiela, 2017), we explored a lightweight Euclidean heuristic that directly enforces hierarchy-consistent distances between subtype representations, without the optimization overhead of a full hyperbolic parameterization (e.g., Riemannian optimization over a Poincaré ball). For each training batch we compute the mean encoder representation (§4.1) of every subtype present in the batch as a prototype, and add a margin-based regularizer that penalizes same-lineage subtype prototypes farther apart than a margin $m_{\text{intra}} = 1.0$ and different-lineage subtype prototypes closer together than a margin $m_{\text{inter}} = 2.0$, added to the training objective (§4.4) with weight 0.1.

Under the single patient-independent split used for architecture development (§3.2), this regularizer did not improve performance and in fact reduced it relative to the same configuration without the term: fine-subtype macro-F1 decreased from 0.171 to 0.136, and coarse-lineage macro-F1 decreased from 0.360 to 0.317. We did not conduct a hyperparameter sweep over the margin values or loss weight, so this diagnostic cannot determine whether the result reflects redundancy, optimization conflict, or an unsuitable configuration. We report the negative single-split result rather than omitting it, but do not treat it as a component ablation or evidence for the necessity of any retained DropCascade component.

A.6 Coarse–Fine Consistency Diagnostic

Independent coarse and fine classifiers can disagree about the hierarchy even when each prediction is plausible in isolation. Table 8 reports their natural agreement. Unlike an earlier single-split analysis, this version uses the same five-fold pooled out-of-fold prediction artifacts as the main hierarchy results, so every patient contributes exactly once to both evaluations.

Figure 2: Patient-balanced lineage-resolved distributions for the three highest-ranked stable PSAS aptamers. Top: raw patient-lineage means. Bottom: means after subtracting each patient's per-aptamer median. Each dot is one patient and boxes summarize patients, not cells; group sizes are shown on the horizontal axis. Corresponding raw and centered panels share the same vertical scale. Aptamer selection and visualization use the full cohort and are exploratory.

Model	Coarse–Fine Consistency
DropCascade	0.812
Flat FT-style Transformer	0.795
FT-style Transformer + HCE	0.834
XGBoost	0.736
Logistic Regression	0.568
Linear SVM	0.631
Random Forest	0.573
Reference Correlation	0.380

Table 8: Natural coarse–fine consistency on the same 361,792-cell, 40-patient pooled out-of-fold artifacts as Table 2. For each model, the hard fine prediction is mapped to its parent and compared with the corresponding hard coarse prediction. *Unknown* is excluded. Consistency is supporting evidence rather than predictive superiority because it can be changed by constrained decoding.

Outer Fold	δ	B^*	LR Macro-F1	XGBoost Macro-F1
0	0.02	10	1.000	1.000
1	0.02	10	1.000	1.000
2	0.02	10	1.000	1.000
3	0.02	10	1.000	1.000
4	0.02	10	1.000	1.000
All folds	**0.02**	**10**	**1.000**	**1.000**

Table 9: Nested patient-level panel selection for disease recognition. In each outer fold, aptamer ranking and budget selection are performed using outer-training patients only, and the selected budget is evaluated once on the untouched outer-test patients. Under the prespecified tolerance margin $\delta = 0.02$, all five folds select the same development budget $B^* = 10$, and the resulting 10-aptamer panels preserve the observed full-panel patient macro-F1 for both Logistic Regression and XGBoost. This result supports a measurement-efficiency interpretation rather than a claim of unique causal biomarkers.

B	Model	Top-B	Random mean [95% interval]	Percentile
5	LR	0.898	0.663 [0.422, 0.890]	97.0
5	XGBoost	0.925	0.725 [0.453, 0.890]	99.0
10	LR	1.000	0.868 [0.670, 0.952]	100.0
10	XGBoost	1.000	0.921 [0.781, 1.000]	97.0
20	LR	1.000	0.973 [0.900, 1.000]	79.5
20	XGBoost	1.000	0.981 [0.923, 1.000]	75.0

Table 10: Compact-panel random controls over 100 independently sampled panels. Top-B uses a fold-specific LR ranking fitted only on outer-training patients. The percentile uses the empirical midrank of Top-B within the random distribution.

A.7 Minimal Aptamer Panel for Disease Classification

Because a smaller aptamer panel would be cheaper and simpler to deploy clinically, we asked how many of the 293 aptamers are actually needed to preserve the observed within-cohort disease result. We therefore performed a nested patient-level feature-selection analysis in which aptamer ranking and budget selection are carried out using outer-training patients only, while outer-test patients are never used for ranking or for choosing the final budget. Within this protocol, all five outer folds select the same development tolerance budget, $B^* = 10$, under a prespecified $\delta = 0.02$ criterion relative to the full panel.

The resulting development-selected 10-aptamer panels reproduce the full-panel patient-level macro-F1 of 1.0 for both Logistic Regression and XGBoost on all five outer test folds. Figure 1 compares these selected panels with 100 independently sampled Random-B controls at each compact budget. For every model and budget, we report the random-panel mean, empirical 2.5th–97.5th percentile interval, and the empirical midrank percentile of Top-B within that distribution (Table 10).

At $B = 10$, the Top-B panel is at the 100th empirical percentile for LR and the 97th for XGBoost; Random-B means are 0.868 and 0.921, respectively. By $B = 20$, random means rise to 0.973 and 0.981 and many random panels also saturate, reducing the corresponding midrank percentiles to 79.5 and 75.0. Thus ranking is most consequential at the smallest budgets, while the broader panel contains extensive substitutable signal.

We next crossed aptamer budget $B \in \{5, 10, 20, 293\}$ with equal-cell caps $C \in \{100, 500, 1000, 5000, \text{all}\}$. For each finite cap, cells were sampled without replacement within each outer-test patient; patient majority votes were pooled across the five held-out folds, and the process was repeated over 50 sampling seeds. This is a sensitivity analysis of held-out predictions rather than a new independently trained cohort. At $B = 5$, macro-F1 varies with the sampled cells and reaches 0.881 on average at $C = 100$; at $B = 10$, 20, and 293, macro-F1 is 1.0 for every cap and every seed (Table 11).

Finally, we tested whether the compact result depends on Logistic Regression coefficients as the attribution mechanism. Within each outer-training fold, we fitted a full-panel XGBoost model and

B	100 cells	500 cells	1,000 cells	5,000 cells	All cells
5	0.881 [0.845, 0.925]	0.894 [0.852, 0.925]	0.906 [0.875, 0.925]	0.899 [0.898, 0.898]	0.898 [0.898, 0.898]
10	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]
20	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]
293	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]

Table 11: LR patient-level macro-F1 under equal-cell subsampling. Entries are mean [empirical 95% interval] over 50 sampling seeds; predictions are pooled across the five outer test folds before scoring the 40 patients.

B	LR–SHAP Jaccard	Ranking	LR eval.	XGB eval.
10	0.501 ± 0.126	LR coefficient	1.000	1.000
10	–	XGBoost-SHAP	1.000	1.000
20	0.475 ± 0.087	LR coefficient	1.000	1.000
20	–	XGBoost-SHAP	1.000	1.000

Table 12: Cross-model attribution control. Jaccard is mean $\pm$ SD across five outer-fold Top-B sets. Evaluation columns report pooled patient-level macro-F1 when the selected panel is used by LR or XGBoost.

computed absolute TreeSHAP contributions on at most 1,000 cells per training patient. We first averaged over cells within each patient and then over patients, preventing high-cell-count patients from dominating the attribution ranking. The LR–SHAP Jaccard overlap is moderate rather than complete (0.501 for Top-10 and 0.475 for Top-20), yet panels from either ranking achieve patient macro-F1 1.0 when evaluated with either LR or XGBoost (Table 12). This cross-model control supports measurement efficiency and predictive redundancy, not a unique molecular signature.

The fold-overlap LR ranking is also stable. The highest-ranked aptamers by mean outer-fold rank are APT-179, APT-225, APT-164, APT-166, APT-57, APT-69, APT-125, APT-161, APT-172, and APT-218, and 290 of the 293 aptamers are retained after a simple redundancy screen at $|r| \leq 0.9$. Appendix A.4 visualizes the top three at the patient-lineage level, including a patient-centering control; because only anonymized IDs are available, neither analysis supports protein-target or pathway interpretation.

A.8 Balanced Sampling Ablation

The final DropCascade configuration (§4.4) does not use class-balanced sampling. We initially implemented a balanced subtype-disease batch sampler to counteract the long-tailed subtype distribution (§3.3), but found and fixed an implementation issue in which `steps_per_epoch` was computed from the natural data loader's batch size (512) rather than the sampler's own, much smaller, per-batch cell count (e.g., 14 subtypes $\times$ 2 diseases $\times$ 4 cells $=$ 112); this mismatch caused each nominal epoch to cover only about 22% of the training set when the sampler was enabled, silently under-training the model and desynchronizing every epoch-indexed schedule (warmup, contrastive ramp-up, routing ramp-up, early-stopping patience) relative to the sampler-disabled baseline.

After fixing this issue, the balanced sampler improved fine-subtype macro-F1 from 0.102 to 0.119 on the single patient-independent split used for architecture development, but both remain below the 0.171 achieved without any balanced sampling at all. We report this ablation as a diagnostic finding about our specific sampler implementation, not as evidence against balanced sampling in general, and did not further redesign the sampler given this result; the final configuration therefore omits it.

A.9 Exploratory Disease-Level Novelty Track

We treat disease-level novelty as an exploratory appendix track rather than a main contribution. In this protocol, all patients from one disease category are withheld during training and evaluated as a disease-level distribution shift. This is useful for stress-testing representations, but it is not a fair basis for method ranking because the cohort contains only six disease groups, proxy-disease supervision is limited, and the supervised detector and training-free confidence scores do not receive matched novelty supervision.

Held-out Disease	Fine Macro-F1	Fine Acc.	Coarse Macro-F1	Coarse Acc.	Tail Macro-F1
BTC	0.086	0.317	0.310	0.479	0.073
CRC	0.105	0.277	0.309	0.520	0.112
GC	0.063	0.269	0.281	0.488	0.057
HCC	0.063	0.246	0.264	0.383	0.072
PC	0.057	0.253	0.221	0.482	0.037
Macro-average	**0.075**	**0.272**	**0.277**	**0.470**	**0.070**

Table 13: Leave-one-disease-out generalization for DropCascade over the five malignant held-out diseases. Each row withholds all patients of the named disease from training. The in-distribution reference using the same model under the standard patient-disjoint 5-fold protocol is fine Macro-F1 0.152 and coarse Macro-F1 0.326. A separate Normal-held-out run produces near-chance fine performance. LODO is reported only as an exploratory appendix stress test, not as proof that the model already solves cross-disease generalization.

Held-out Disease	Proxy-Sup. Head AUROC	Proxy-Sup. Head AUPR	Energy AUROC
BTC	0.782	0.855	0.591
CRC	0.874	0.901	0.779
GC	0.729	0.698	0.575
HCC	0.881	0.842	0.558
Normal	0.607	0.843	0.722
PC	0.999	0.999	0.539
Macro-average	**0.812**	**0.856**	**0.627**

Table 14: Exploratory novelty detection under the nested leave-one-disease-out protocol. For each held-out disease, the proxy-supervised head is learned on a proxy-novel disease drawn from the remaining training diseases and then evaluated on the truly held-out disease, which is unseen during both cascade and novelty-head training. Energy is a training-free confidence score and does not receive matched proxy-novel supervision, so this table is descriptive rather than a fair method comparison. AUPR additionally depends on the held-out-disease prevalence in each evaluation set.

Table 14 reports two standard training-free scores, maximum softmax probability (Hendrycks & Gimpel, 2016) and energy (Liu et al., 2020), alongside a proxy-supervised novelty head. For each held-out disease, the cascade is trained only on the known diseases. The encoder is then frozen, and a lightweight MLP novelty head is trained to distinguish one proxy-novel known disease from the remaining known diseases. The truly held-out disease is never used during either cascade training or novelty-head training.

Under this exploratory nested leave-one-disease-out protocol, the proxy-supervised head attains higher descriptive AUROC than the training-free confidence scores. This should be interpreted narrowly: proxy disease supervision can help separate a withheld disease from known diseases in this cohort. It does not establish that the novelty head is intrinsically better than energy scoring, because the comparison lacks matched supervised probes on raw APT features, classical classifiers, patient-clustered uncertainty intervals, proxy-choice sensitivity, and a formal ID/OOD patient manifest.